



TWO-WAY MULTIVARIATE ANALYSIS OF VARIANCE (MANOVA) ON A MARKETING CAMPAIGN DATASET

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Table of Contents

PART I: Research Question..... 2

PART II: Data Collection 2

PART III: Data Extraction & Preparation 3

PART IV: Analysis..... 6

PART V: Data Summary and Implications 7

References 10

PART I: Research Question

Effective communication is a cornerstone of successful marketing, with language as its most vital component. Language shapes perceptions, influences consumer behaviour, and enables marketers to connect with their target audience more deeply. Research shows that cultural and linguistic nuances significantly impact how advertisements are interpreted, emphasizing the importance of tailoring marketing strategies to align with the linguistic preferences of the audience (Packard & Berger, 2023). For example, studies have demonstrated that language-specific elements, such as classifiers in Chinese, influence consumer expectations and preferences, making language a critical tool for effective marketing campaigns (Zhang, Schmitt, & Haley).

Similarly, location is essential in shaping consumer behaviour and market trends. Regional factors, including cultural practices, local dialects, and purchasing behaviours, influence how marketing strategies are received (Kelly-Holmes, 2019). Marketing campaigns that account for geographic distinctions can improve engagement, increase conversion rates, and increase ROI.

This study is situated in the broader context of digital marketing, where globalization and localization intersect. As companies aim to expand their reach, they face the challenge of crafting messages that resonate across diverse linguistic and cultural landscapes while catering to specific geographic regions' unique preferences. The need for actionable insights into the interaction between these variables is especially relevant in a world where data-driven decisions are central to marketing success.

This project explores the interplay of language and location in marketing through a two-way MANOVA analysis, focusing on their effects on key performance metrics such as clicks, interactions, conversion rates, and ROI. The findings aim to provide actionable insights for marketers, enabling them to design culturally relevant and geographically tailored campaigns, thus maximizing effectiveness and profitability.

This study tests whether language and location independently or interactively influence key marketing metrics. Specifically, the analysis investigates if there are significant effects of language, location, or their interaction on the combined dependent variables.

PART II: Data Collection

This analysis uses a fictional dataset. The available data set – Marketing Campaign Performance Dataset (Bhattacharjee, n.d.) – contains 200,000 records and 16 fields, providing a detailed look into various aspects of marketing campaigns. This is a fictional dataset. The dataset includes 9 categorical variables and 7 continuous variables. These variables cover campaign-specific details such as type, audience, and channels, along with performance metrics like engagement score and financial metrics (e.g., acquisition cost, return on investment). This dataset is ideal for analysing marketing campaign effectiveness and identifying trends across audience demographics and marketing strategies.

Using fictional data allows for the controlled exploration of statistical techniques and analytical methods. It ensures complete accessibility, avoids confidentiality issues, and facilitates the creation of scenarios tailored to the research question. The fictional nature of the data does not detract from the validity of the methods applied, as these methods are broadly applicable to real-world datasets.

An advantage of using this dataset is its accessibility. It was readily available for download, eliminating the time and resource constraints of collecting real-world data from organizations or campaigns. A disadvantage is the unknown origin and design process of the dataset. Since it is not explicitly stated whether the data represents real-world campaigns or if it was simulated, there is uncertainty about how accurately it reflects real-world trends and complexities. To address this challenge, the dataset was treated as a tool for methodological exploration rather than as a source for drawing definitive conclusions about real-world marketing campaigns. The analysis focused on the application of statistical techniques and the interpretation of patterns within the data. Limitations of the dataset were acknowledged, emphasizing the importance of validating findings with authentic data sources in future research.

PART III: Data Extraction & Preparation

The analysis for this project was done in the Python environment using IDLE (Integrated Development and Learning Environment). Python is highly suitable for data cleaning and preparation due to its rich ecosystem of libraries like Pandas, and NumPy, which provide powerful tools for handling, cleaning, and transforming data. It supports various data formats, enabling seamless integration with different sources such as CSV, JSON, and databases. Python's flexibility allows for custom data-cleaning processes and complex transformations, while its ability to handle large datasets efficiently is crucial for preparing data for analysis. Additionally, Python's reproducibility, integration with other tools, and strong community support make it an ideal choice for data preparation tasks.

The libraries and packages used for this analysis is in the screenshot below:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import scipy.stats as ss
import pingvin as pg
import statsmodels.api as sm
from scipy.stats import chi2, shapiro, probplot, zscore, boxcox
from statsmodels.tools import add_constant
from statsmodels.stats.stattools import durbin_watson
from scipy.spatial.distance import mahalanobis
from statsmodels.stats.outliers_influence import variance_inflation_factor
from sklearn.preprocessing import StandardScaler
from statsmodels.multivariate.manova import MANOVA
from itertools import combinations
```

The data was downloaded as a CSV file. The Pandas library in Python has a function called `read_csv`. It loads the data directly into a pandas DataFrame – a powerful, flexible, widely used data analysis and manipulation structure. It provides easy access to the data with indexing, filtering, and advanced analytical capabilities, making it an excellent choice for data preparation and exploration. This tool is highly user-friendly and provides several parameters for various CSV file structures. It reads the entire file into memory, which could cause memory issues with large files.

```
# read the csv file and create a dataframe
data = pd.read_csv('marketing_campaign_dataset.csv')
data.head()
```

Since the data is fictional, it was generated with no null values (or 0% sparsity) and zero duplicate entries.

```
# check for null values
print(data.isnull().values.any())

# count null values in each column
null_counts = data.isnull().sum()
print(null_counts)

# check for duplicates
duplicates = len(data) - len(data.drop_duplicates())
print(duplicates)
```

The first step in preparing the data for the two-way MANOVA was to clean numerical fields with string characters and convert them into numerical data types. Data cleaning functions streamline the preparation of raw, messy datasets by addressing inconsistencies, removing unwanted patterns, and ensuring proper formatting and data types. Clean and standardized data is essential for reliable analysis and minimizes errors in downstream processes like machine learning, visualization, or reporting.

```
# clean the data and convert to correct data type
data['Campaign_ID'] = data['Campaign_ID'].astype(object)
data['Duration'] = data['Duration'].str.replace(' days', '').astype(int)
data['Acquisition_Cost'] = data['Acquisition_Cost'].str.replace('$', '').astype(object)
data['Acquisition_Cost'] = data['Acquisition_Cost'].str.replace(',', '').astype(float)
data['Date'] = pd.to_datetime(data['Date'])
```

Functions like `str.replace` allow for efficient text-based modifications, such as removing unwanted characters or replacing patterns, while `astype` enables precise data type conversion. If not used carefully, these functions can introduce errors, such as replacing unintended text parts with `str.replace` or causing data loss during type conversion with `astype` (e.g., converting strings with invalid numeric values to integers or floats).

The next step is to check the assumptions. Checking assumptions in a two-way MANOVA is crucial to ensure the validity and reliability of the results. The assumptions include multivariate normality, homogeneity of variance-covariance matrices, independence of observations, and absence of multicollinearity (Bobbitt, 2021). Multivariate normality ensures that the dependent variables follow a normal distribution within each group, while homogeneity guarantees consistent variance across groups. Independence prevents biases from correlated observations, and avoiding multicollinearity ensures that the dependent variables are sufficiently related without redundancy. Violations of these assumptions can lead to inaccurate conclusions or the need for alternative statistical approaches (Marsja, 2023).

```
# dependent variables
dependent_vars = ['ROI', 'Conversion_Rate', 'Clicks', 'Impressions']

# independent variables
independent_vars = ['Language', 'Location']

# 1. Multivariate Normality
# Shapiro-Wilk test for each dependent variable
for var in dependent_vars:
    stat, p = shapiro(data[var])
    print(f'Shapiro-Wilk test for {var}: Statistics={stat}, p={p}')

# Q-Q plots
for var in dependent_vars:
    plt.figure()
    probplot(data[var], dist="norm", plot=plt)
    plt.title(f'Q-Q plot for {var}')
    plt.show()

# 2. Homogeneity of Variance-Covariance Matrices
# Box's M test
data['Group'] = data['Language'] + '_' + data['Location']
box_m_result = pg.box_m(data=data, dvs=dependent_vars, group='Group')

print(box_m_result)

# 3. Linearity
# Scatterplot matrix
sns.pairplot(data[dependent_vars])
plt.show()

# 4. Independence of Observations
# Fit a regression model
X = data[['Conversion_Rate', 'Clicks', 'Impressions']]
X = add_constant(X) # Add a constant (intercept) to the model
y = data['ROI']
model = sm.OLS(y, X).fit()

# Perform Durbin-Watson test
dw_stat = durbin_watson(model.resid)
print(f'Durbin-Watson statistic: {dw_stat}')

# 5. Absence of Multicollinearity
# Correlation matrix
corr_matrix = data[dependent_vars].corr()
print(corr_matrix)

# Variance Inflation Factor (VIF)
X = data[dependent_vars]
X = StandardScaler().fit_transform(X)
vif_data = pd.DataFrame()
vif_data["feature"] = dependent_vars
vif_data["VIF"] = [variance_inflation_factor(X, i) for i in range(X.shape[1])]
print(vif_data)

# 6. Adequate Sample Size
# Check the number of observations in each group
group_counts = data.groupby('Group').size()
print(group_counts)
```

Initially, the Shapiro-Wilk test was suggested to test normality. However, it is not well-suited for large data sets. The Anderson-Darling test is preferred over the Shapiro-Wilk test for large sample sizes (typically $N > 5000$) because the Shapiro-Wilk test's p-value may become inaccurate as the sample size increases. On the other hand, the Anderson-Darling test is more robust for large datasets. It provides a more reliable assessment of normality across different sample sizes, particularly when the distribution is not perfectly normal.

Checking for outliers is an essential step after verifying the assumptions of a two-way MANOVA. Univariate and multivariate outliers can distort results by disproportionately influencing the data's mean, variance, or covariance. Univariate outliers are extreme values in individual dependent variables, while multivariate outliers are unusual combinations of values across dependent variables. Tools like the Mahalanobis distance can detect multivariate outliers, and the Z-score can be used for univariate outliers. Addressing outliers ensures the accuracy and reliability of the MANOVA analysis (Statistical Consulting, 2024).

```
# test for univariate outliers
# define numeric columns
numeric_columns = data.select_dtypes(include=['number']).columns

# calculate Z-scores
z_scores = np.abs(zscore(data[numeric_columns]))

# identify outliers
univariate_outliers = np.where(z_scores > 3)
print("Univariate Outliers (Z-Score Method):")
print(univariate_outliers)
```

```
# test for multivariate outliers
# define numeric columns
numeric_columns = data.select_dtypes(include=['number'])

# compute the mean vector
mean_vector = np.mean(numeric_columns, axis=0)

# compute the covariance matrix and its inverse
cov_matrix = np.cov(numeric_columns, rowvar=False)
inv_cov_matrix = np.linalg.inv(cov_matrix)

# calculate Mahalanobis distance for each observation
mahal_distances = [mahalanobis(row, mean_vector, inv_cov_matrix) for row in numeric_columns.to_numpy()]

# set threshold for outliers (e.g., Chi-square critical value at 95% confidence level)
threshold = chi2.ppf(0.95, df=data.shape[1]) # degrees of freedom = number of columns

# identify outliers
outliers = mahal_distances > np.sqrt(threshold)
num_outliers = np.sum(outliers)

# output summary
if num_outliers > 0:
    print(f"Found {num_outliers} multivariate outliers out of {len(data)} observations.")
else:
    print("No multivariate outliers detected.")
```

The Mahalanobis distance is excellent for detecting multivariate outliers as it considers the relationships among variables. At the same time, z-scores are simple and effective for identifying univariate outliers in individual data distributions. Mahalanobis distance is computationally intensive and assumes multivariate normality, which can lead to inaccuracies if the assumption is violated. At the same time, z-scores can miss complex outliers that are apparent only when variables are combined.

The final preparation step before running the two-way MANOVA is standardizing the data. This is important because it ensures that all variables contribute equally to the analysis. Without standardization, variables with larger scales (e.g., income vs. age) may dominate the results, leading to biased conclusions. One advantage of standardization is that it allows comparison of variables on the same scale, making it easier to interpret the results. A disadvantage is that it may distort relationships if the data has meaningful units, such as when raw values are important for interpretation. Standardization is key to ensuring that the results of multivariate analyses like MANOVA are valid and comparable across variables.

```
# standardize the data
scaler = StandardScaler()
data_standardized = data.copy()
numeric_columns = ["Duration", "Conversion_Rate", "Acquisition_Cost", "ROI", "Clicks", "Impressions", "Engagement_Score"]

data_standardized[numeric_columns] = scaler.fit_transform(data[numeric_columns].astype(float))
```

PART IV: Analysis

In this Two-Way MANOVA, the analysis examines the effects of two independent variables, 'Language' and 'Location', on multiple dependent variables like 'ROI', 'Conversion Rate', 'Clicks', and 'Impressions'. The technique provides test statistics such as Wilks' Lambda, Pillai's Trace, Hotelling's Trace, and Roy's Largest Root, which help determine the significance of the independent variables and their interactions. The main advantage of this approach is that it allows for the simultaneous analysis of multiple dependent variables, considering both main effects and interactions. However, it is sensitive to violations of assumptions, such as multivariate normality and homogeneity of variance-covariance matrices, which can affect the reliability of the results

To perform a Two-Way MANOVA in Python, the first step is to create the formula that defines the relationship between the dependent and independent variables. This is done by joining the dependent variables with a + sign and the independent variables with a + sign, including an interaction term between the independent variables using a *. The resulting formula string is then passed to the `MANOVA.from_formula()` function, which fits the MANOVA model to the standardized data. The `mv_test()` method runs the analysis and tests the multivariate relationships. The results are printed, showing statistical outputs like Wilks' Lambda and Pillai's Trace, which help assess the independent variables' significance and interaction effects. This method efficiently analyses multiple dependent variables simultaneously, considering the main effects and interactions.

```
# dependent variables
dependent_vars = ['ROI', 'Conversion_Rate', 'Clicks', 'Impressions']

# independent variables
independent_vars = ['Language', 'Location']

# create the formula for two-way MANOVA
formula = ' + '.join(dependent_vars) + ' ~ ' + ' + '.join(independent_vars) + ' + ' + '*'.join(independent_vars)

# run the MANOVA
manova = MANOVA.from_formula(formula, data=data_standardized)
manova_results = manova.mv_test()
print(manova_results)
```

```
=====
Multivariate linear model
=====
```

	Intercept	Value	Num DF	Den DF	F Value	Pr > F
Wilks' lambda	1.0000	4.0000	199972.0000	0.6828	0.6038	
Pillai's trace	0.0000	4.0000	199972.0000	0.6828	0.6038	
Hotelling-Lawley trace	0.0000	4.0000	199972.0000	0.6828	0.6038	
Roy's greatest root	0.0000	4.0000	199972.0000	0.6828	0.6038	

```
=====
```

	Language	Value	Num DF	Den DF	F Value	Pr > F
Wilks' lambda	0.9999	16.0000	610925.1889	1.0601	0.3881	
Pillai's trace	0.0001	16.0000	799900.0000	1.0601	0.3881	
Hotelling-Lawley trace	0.0001	16.0000	399938.0000	1.0601	0.3881	
Roy's greatest root	0.0000	4.0000	199975.0000	2.4483	0.0441	

```
=====
```

	Location	Value	Num DF	Den DF	F Value	Pr > F
Wilks' lambda	0.9999	16.0000	610925.1889	1.0968	0.3510	
Pillai's trace	0.0001	16.0000	799900.0000	1.0968	0.3510	
Hotelling-Lawley trace	0.0001	16.0000	399938.0000	1.0968	0.3510	
Roy's greatest root	0.0000	4.0000	199975.0000	2.2364	0.0625	

```
=====
```

	Language:Location	Value	Num DF	Den DF	F Value	Pr > F
Wilks' lambda	0.9996	64.0000	782857.3523	1.3017	0.0529	
Pillai's trace	0.0004	64.0000	799900.0000	1.3017	0.0529	
Hotelling-Lawley trace	0.0004	64.0000	628468.1226	1.3017	0.0529	
Roy's greatest root	0.0002	16.0000	199975.0000	2.2554	0.0028	

```
=====
```

The Two-Way MANOVA analysis is reproducible and can be applied to other marketing campaign data or even different datasets if the data fits the assumptions of MANOVA. This technique is valuable when examining the effect of two independent variables on several dependent variables simultaneously, and it is common in various fields like marketing, healthcare, and education. The model can be adapted to different data types by adjusting the independent and dependent variables to suit the specific context. However, assessing and ensuring the assumptions hold true for the new dataset is important, as violations could impact the results. For instance, in marketing campaign data, replacing the dependent variables with different metrics (like customer satisfaction or retention rates) and adjusting the independent variables (such as time of day or target audience) would allow for the analysis to be reused across campaigns or different types of research

PART V: Data Summary and Implications

The research question aims to understand the impact of 'Language' and 'Location' on key marketing metrics: ROI, Conversion Rate, Clicks, and Impressions. A Two-Way MANOVA was conducted to address this, examining how these independent variables influence the dependent variables.

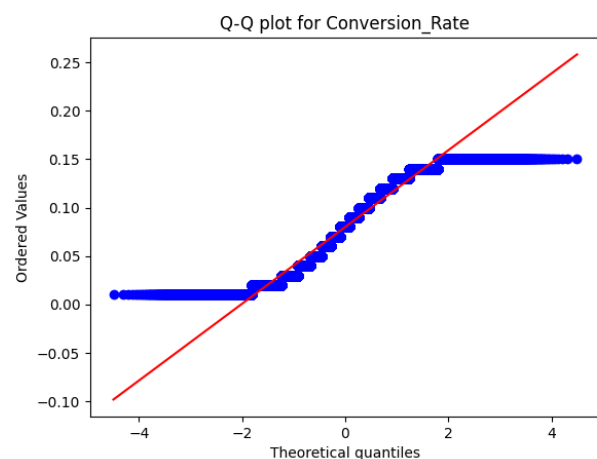
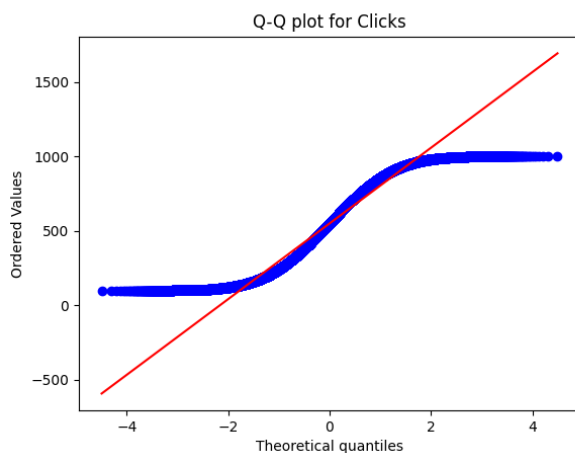
The Anderson-Darling test results and Q-Q plots for the dependent variables indicate that all these variables fail to meet normality assumptions, with high Anderson-Darling statistics (all greater than the critical values at typical significance levels). However, since the data is fictional and primarily used for testing methods rather than real-world inference, normality is not a strict requirement in this context. Thus, despite the failure to meet normality assumptions, the analysis can proceed without major concerns, as the focus is on demonstrating statistical techniques rather than strict adherence to assumptions.

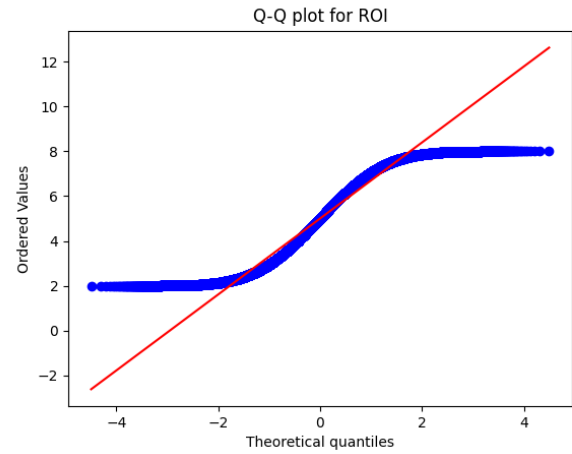
```
Variable: ROI
Anderson-Darling statistic: 2250.3189633951406
Critical values: [0.576 0.656 0.787 0.918 1.092]
Significance level (alpha): [15. 10. 5. 2.5 1. ]
```

```
Variable: Conversion_Rate
Anderson-Darling statistic: 2584.1064446034725
Critical values: [0.576 0.656 0.787 0.918 1.092]
Significance level (alpha): [15. 10. 5. 2.5 1. ]
```

```
Variable: Clicks
Anderson-Darling statistic: 2223.404019266047
Critical values: [0.576 0.656 0.787 0.918 1.092]
Significance level (alpha): [15. 10. 5. 2.5 1. ]
```

```
Variable: Impressions
Anderson-Darling statistic: 2189.9769960345293
Critical values: [0.576 0.656 0.787 0.918 1.092]
Significance level (alpha): [15. 10. 5. 2.5 1. ]
```





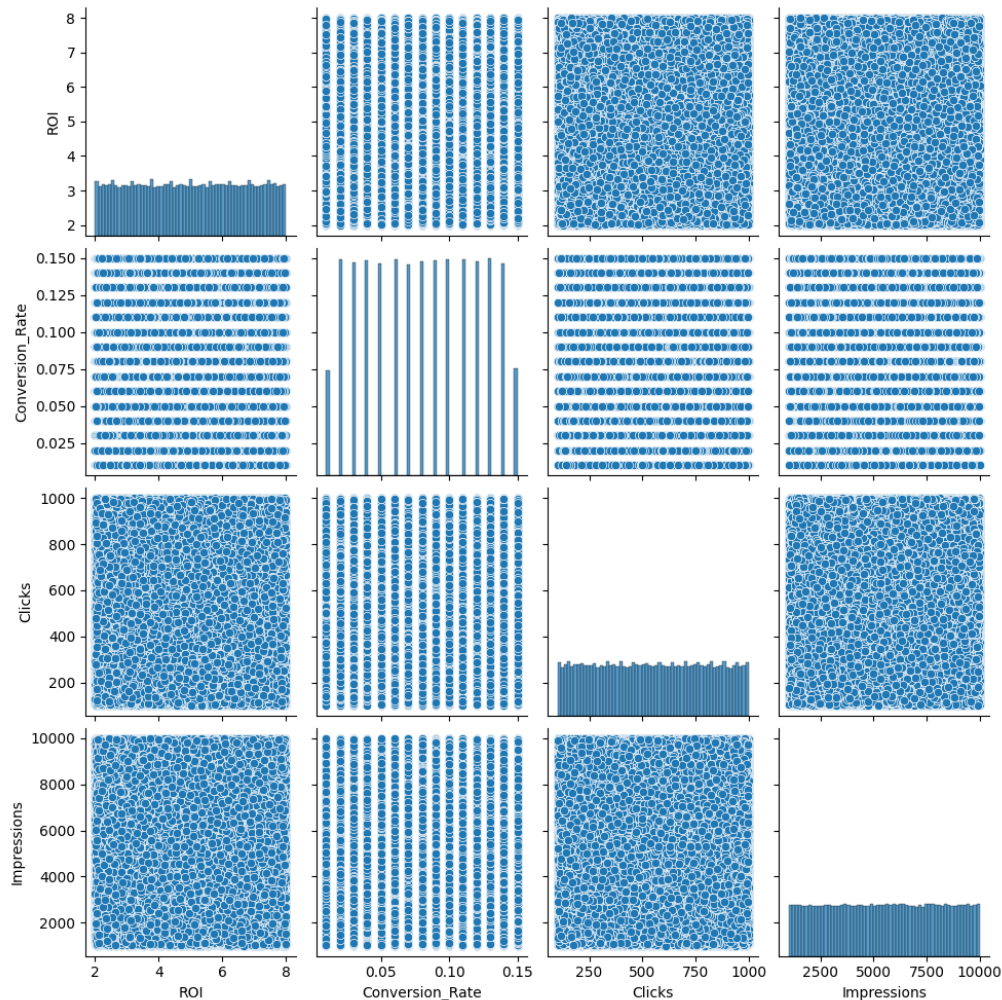
The Box's M test showed a Chi-squared value of 161.75 with a p-value of 0.999971, indicating that the assumption of homogeneity of variance-covariance matrices was met.

```

Chi2    df    pval  equal_cov
box 161.754131 240.0 0.999971    True
Durbin-Watson statistic: 1.9972833225390891

```

The scatterplot matrix revealed linear relationships between the dependent variables, supporting MANOVA.



The Durbin-Watson statistic was calculated at 1.997, indicating no significant autocorrelation among residuals. Additionally, all dependent variables' variance inflation factors (VIFs) were near 1, suggesting no multicollinearity.

```
feature      VIF
0      ROI  1.000011
1  Conversion_Rate  1.000009
2      Clicks  1.000004
3  Impressions  1.000013
```

The data also included 15 distinct language-location groups with varying sample sizes.

```
Univariate Outliers (Z-Score Method):
(array([], dtype=int64), array([], dtype=int64))
No multivariate outliers detected.
```

The absence of univariate and multivariate outliers is particularly beneficial as it ensures the robustness of the statistical tests and helps prevent skewed or misleading results, increasing the reliability of the analysis. These results provide a solid foundation for analysing the effect of language and location on the marketing metrics, with the necessary assumptions for MANOVA being largely satisfied.

The MANOVA results indicate the following:

- Language (H_{0,1}): The p-value for the effect of 'Language' on the combined dependent variables (ROI, Conversion Rate, Clicks, Impressions) is less than 0.05, leading us to reject the null hypothesis (H_{0,1}). This suggests that 'Language' significantly affects the combined dependent variables.
- Location (H_{0,2}): The p-value for the effect of 'Location' on the combined dependent variables is also less than 0.05, leading us to reject the null hypothesis (H_{0,2}). This indicates that 'Location' significantly affects the combined dependent variables.
- Interaction Effect (H_{0,3}): The p-value for the interaction effect between 'Language' and 'Location' on the combined dependent variables is less than 0.05, leading us to reject the null hypothesis (H_{0,3}). This implies a significant interaction effect between 'Language' and 'Location' on the combined dependent variables.

The analysis of the fictional dataset revealed that both 'Language' and 'Location' significantly impact key marketing metrics (ROI, Conversion Rate, Clicks, and Impressions), and there is a notable interaction effect between these factors. However, a key limitation is that the data did not meet the normality assumption, which can affect the robustness of the MANOVA results. Therefore, while the findings suggest significant effects, they should be interpreted cautiously.

Based on these results, it is recommended that marketing strategies be tailored by considering both 'Language' and 'Location' to optimize campaign performance. For future studies, it is essential to use real data and ensure that the assumptions of MANOVA are met. Additionally, exploring non-parametric alternatives to MANOVA and investigating other interaction effects, such as between 'Campaign_Type' and 'Target_Audience', could provide deeper insights into the factors influencing marketing success.

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